

ANWESHA BHATTACHARJEE

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Education

University of Glasgow, Glasgow, UK

Sept 2024 – Oct 2025

MSc in Astrophysics – Awarded with Merit

Gargi College, University of Delhi, Delhi, India

Oct 2021 – June 2024

BSc. Physics (Hons) – GPA: 8.824/10

Research Experience

Research Project (Remote)

May 2026 – Ongoing

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

Supervisor: Dr. Manasadevi P Thirugnanasambandam

- Investigating optical sensing of a three-mirror coupled cavity for gravitational-wave detector applications.
- Developing numerical models using **finesse** to study cavity sensing, optical response, and power build-up.
- Analysing longitudinal and angular sensing using PDH error signals and sensing matrices.
- Investigating the affect of cavity parameters, RF modulation frequency, and demodulation phase on sensing performance.

MSc Thesis — Laser Interferometry for Gravitational Wave Detection

June – Sept 2025

University of Glasgow

Supervisor: Dr. Andrew Spencer

- **Experimental Setup:** Designed, built and aligned a tabletop, open-loop Michelson interferometer setup with PZT-driven mirror mount, and photodetector to optimise fringe contrast.
- **Data Acquisition:** Characterised optical fringe signals, photodetector response, and displacement sensitivity using Moku:Go software; analysed the environmental disturbances coupling into the interferometric output by plotting the data acquired on Python.
- **Noise Modelling:** Developed aLIGO noise budgets using **pygwinc**, reproducing published sensitivity curves; performed parameter studies of laser power, arm length, and optical configurations, evaluating their impact on detector sensitivity.
- **Optical Simulation:** Simulated the tabletop Michelson interferometer configurations using **finesse**, analysing beam splitter ratios and fringe visibility under realistic optical constraints.

Research Workshops

Gravitational Wave Detector Modelling & Controls Workshop

July 13–19, 2026

In-person workshop organised by IUCAA, LIGO-India & HSF-India

Instructions by Prof. Rana X. Adhikari (Caltech), Dr. Evan D. Hall (MIT), Prof. Denis Martynov (University of Birmingham)

- Studied transfer functions, feedback control, and suspension dynamics through simulation exercises based on Advanced LIGO detector subsystems.
- Simulated suspension damping and examined the trade-off between damping performance and sensor noise in closed-loop control systems.
- Derived the Pound–Drever–Hall (PDH) error signal and implemented cavity locking using simplified interferometer models.
- Used **pygwinc** to investigate the contributions of quantum, thermal, seismic, suspension, and sensing noise to interferometer sensitivity.
- Explored detector commissioning concepts through guided exercises integrating optical cavities, sensing, and feedback control representative of Advanced LIGO and LIGO-India.

Gravitational Wave Detector Instrumentation Workshop

June 28 – July 11, 2026

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

Hands-on Supervision by Dr. Suresh Doravari, Dr. Manasadevi Thirugnanasambandam, Dr. Ravi Kesharwani, Dr. Shivaraj Kandhasamy, and Dr. Saravanan T. R.

- **Suspension systems:** Characterised a HAUX single-stage suspension across multiple degrees of freedom by analysing sensor data in Python, computing displacement Amplitude Spectral Densities (ASDs) using Welch's method, and identifying noise contributions; calibrated an individual Optical Sensor and Electro-Magnetic Actuator (OSEM) for displacement readout.
- **Seismic isolation:** Characterised an inverted-pendulum isolation stage and a folded-pendulum using 731A ac-

celerometer, L4C geophone, and an FP inertial sensor. Estimated the resonance frequency and damping ratio by fitting a second-order oscillator model to the measured response, and evaluated coherence between sensor outputs.

- **Fabry–Pérot cavity:** Aligned a two-mirror optical cavity, measured the free spectral range (FSR), linewidth (FWHM), and finesse, and used JamMT to design a mode-matching solution before remeasuring the cavity performance.
- **Michelson interferometer & feedback control:** Characterised and aligned a Michelson interferometer, implemented PID-based lock acquisition using a PZT-actuated mirror, measured fringe visibility, and injected seismic, acoustic, and electronic noise sources to characterise their coupling and coherence with the locked error signal.

Selected Academic Projects

Spectral Analysis of the X1.2 Flare using EIS/Hinode Observations

Jan – April 2025

University of Glasgow — Course: *The Sun's Atmosphere*

- Analysed EIS/Hinode spectroscopic observations of the 15 May 2013 X1.2 solar flare using Python.
- Processed and calibrated FITS data, performed Gaussian line fitting, and generated intensity maps to study plasma dynamics.

Bayesian Parameter Inference

Jan – April 2025

University of Glasgow — Course: *Advanced Data Analysis*

- Implemented Bayesian parameter inference using MCMC methods for linear and quadratic models.
- Performed model comparison and uncertainty quantification through posterior distributions, credible intervals, and Bayes factors.

Technical Skills

Programming & Scientific Computing: Python (NumPy, SciPy, Matplotlib, AstroPy; signal processing using Welch's method), SciLab

Simulation & Modelling: FINESSE (interferometer simulation), pygwinc (gravitational-wave detector noise budgets), JamMT (mode-matching design), Bayesian inference (basic), Markov Chain Monte Carlo (basic)

Data Analysis & Control: Spectral analysis (Welch's method), coherence analysis, Gaussian fitting, second-order oscillator fitting, transfer function analysis, PID feedback control, Pound–Drever–Hall (PDH) sensing and cavity locking (basic)

Experimental Techniques: Laser alignment, Michelson and Fabry–Pérot interferometers, mode matching, photodiode and beam-splitter calibration, displacement calibration, OSEM calibration, HAUX suspension characterisation, inverted-pendulum and folded-pendulum seismic isolation, L-4C geophone instrumentation, Moku:Go

Tools: GitHub, Jupyter Notebook, LaTeX

Selected Presentations & Academic Activities

Contributed Talk, GRASS 2026 Conference

June 2026

Abstract accepted for a contributed oral presentation at the GRASS 2026 Conference on gravitational-wave science and related research topics, hosted by the University of Padua, Italy.

Scholarships

University of Glasgow International Leadership Scholarship

Sept 2024 – Sept 2025

Awarded a £10,000 scholarship by the University of Glasgow in recognition of academic excellence and leadership potential, supporting postgraduate studies in Astrophysics.

Leadership & Engagement

International Student Representative

Sept 2024 – Nov 2025

University of Glasgow

- Represented international MSc student perspectives in institutional panels and outreach events.

Core Member & Leadership Roles

Nov 2021 – June 2024

QUASAR - The Physics Society, Gargi College, University of Delhi

- PR Head, Vice-President, and Joint-Secretary; led organisation and coordination of inter-collegiate academic events with 150+ participants, coordinating teams and external collaborations.

Co-Editor & Proofreader

Jan 2022 – May 2024

Physikos - Physics Departmental Magazine, Gargi College, University of Delhi

- Managed editorial review and publication coordination.